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Natriuretic Peptide Receptor-1 Agonist for Resistant Hypertension

A Randomized Phase 2 Trial

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ABSTRACT

BACKGROUND XXB750 is a potent and long-acting human monoclonal antibody agonist of the natriuretic peptide receptor-1 (NPR-1), the receptor for atrial natriuretic peptide and B-type natriuretic peptide. Activation of NPR-1 may lower blood pressure (BP) in patients with resistant hypertension through increases in the second messenger cyclic guanosine monophosphate.

OBJECTIVES The aim of this study was to determine the effect of NPR-1 agonism on 24-hour BP.

METHODS This phase 2, multicenter, randomized, double-blind, placebo-controlled trial evaluated 4 once-monthly subcutaneous doses of XXB750 (30 mg, 60 mg, 120 mg, and 240 mg) administered at baseline, week 4, and week 8 in patients with resistant hypertension. Resistant hypertension was defined as 24-hour systolic BP (SBP) ≥135 mm Hg despite treatment with 3 or 4 guideline-directed antihypertensive agents. The primary endpoint was the dose-response relationship of XXB750 vs placebo in reducing the mean 24-hour SBP from baseline to week 12.

RESULTS The study population (N = 189) had a mean age of 61 years, was 69% male, 61% White, 20% Asian, and 14% Black/African American, and had a baseline mean 24-hour ambulatory BP of 148/81 ± 11/10 mm Hg. Plasma concentrations of XXB750 dose-dependently increased in the XXB750 groups, and XXB750 dose-dependently increased plasma and urine cyclic guanosine monophosphate concentrations by week 12. However, paradoxically, there was no significant dose-dependent change from baseline in mean 24-hour SBP at week 12. In fact, none of the doses of XXB750 had a greater BP-lowering effect than placebo. Changes in ambulatory BP were not affected by the development of anti-drug antibodies. Although there were no changes from baseline in urinary sodium excretion, there were small increases in plasma renin levels in the XXB750 groups. Adverse events with XXB750 were numerically higher than with placebo. Adherence to background antihypertensive medications, assessed by drug assays in 88 patients with available plasma samples, was consistent across all treatment groups at both baseline and week 12.

CONCLUSIONS XXB750, an NPR-1 agonist, had a neutral effect on 24-hour ambulatory SBP in patients with resistant hypertension, despite showing a robust dose-response relationship for engagement of the target of pharmacologic action as indicated by increases in cyclic guanosine monophosphate levels. (An Efficacy, Safety, Tolerability and Dose Finding Study of XXB750 in Resistant Hypertension Patients; NCT05562934) (JACC. 2026;■:■-■)

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ABBREVIATIONS AND ACRONYMS

ADA = anti-drug antibody

AE = adverse event

ANP = atrial natriuretic peptide

BP = blood pressure

cGMP = cyclic guanosine monophosphate

DBP = diastolic blood pressure

FAS = full analysis set

MANP = mimetic atrial natriuretic peptide

MCP-Mod = multiple comparison procedure-modeling

NPR-1 = natriuretic peptide receptor-1

SAE = serious adverse event

SBP = systolic blood pressure

SC = subcutaneous

Resistant hypertension is defined as blood pressure (BP) that remains above the goal of ≥130/80 mm Hg (U.S. guidelines) and ≥140/90 mm Hg (European guidelines) despite the concurrent use of at least 3 antihypertensive agents of different classes at optimal or maximally tolerated doses, including a diuretic.[1-3] The U.S. guidelines also consider BP that is controlled with ≥4 medications as resistant hypertension.[3]

Patients with resistant hypertension are at a higher risk of cardiovascular and renal events, as well as death.[4] The global prevalence of resistant hypertension among patients with treated hypertension in the United States is approximately 12% and approximately 10% based on the ≥130/80 mm Hg and ≥140/90 mm Hg thresholds, respectively.[4] Identification of the optimal combination of antihypertensive agents to manage hypertension and to ensure long-term adherence to multidrug regimens has been challenging.

Natriuretic peptide receptor-1 (NPR-1) is a membrane-bound receptor that acts as a guanylate cyclase to increase intracellular cyclic guanosine monophosphate (cGMP) levels. Elevated cGMP leads to reductions in BP and plasma volume by promoting vasodilation, increasing renal excretion of sodium and water and suppressing the renin-angiotensin system.[5] XXB750 (a fully human monoclonal IgG1 antibody) is a novel NPR-1 agonist developed to activate all NPR-1 actions that may be beneficial in patients with resistant hypertension. XXB750 is a potent and long-acting NPR-1 agonist with an estimated plasma half-life of approximately 16 days, allowing once-monthly subcutaneous (SC) injection that replicates and enhances the effects of atrial natriuretic peptide (ANP) in preclinical models. In a phase 1 study, XXB750 dose-dependently increased plasma and urinary cGMP (up to 28 days after SC injection) and decreased systolic BP (SBP) by 10 to 25 mm Hg at doses of 30, 60, 120, 240, 450, and 600 mg in study participants who had normal or slightly elevated BP maintained on a regular sodium diet.[6] The BP-lowering effect was at its maximum on day 3 and accompanied by an increase in heart rate. Subsequently, BP returned to baseline levels within 4 to 8 weeks depending on dose.

These first clinical data as well as the BP-lowering effects of M-atrial natriuretic peptide (MANP), a chemically modified ANP mimetic,[7] in patients with primary hypertension and the effects of sacubitril/valsartan (which inhibits neprilysin, the enzyme that breaks down natriuretic peptides) in patients with hypertension, heart failure, and chronic kidney disease[8,9] provided a strong rationale to evaluate this NPR-1 agonist in patients with treatment-resistant hypertension. Hence, we designed a phase 2 study to evaluate the antihypertensive effects, pharmacology, safety, and tolerability of 4 doses of XXB750 (30 mg, 60 mg, 120 mg, and 240 mg) administered once every 4 weeks as SC injections in patients with resistant hypertension confirmed by ambulatory BP monitoring.

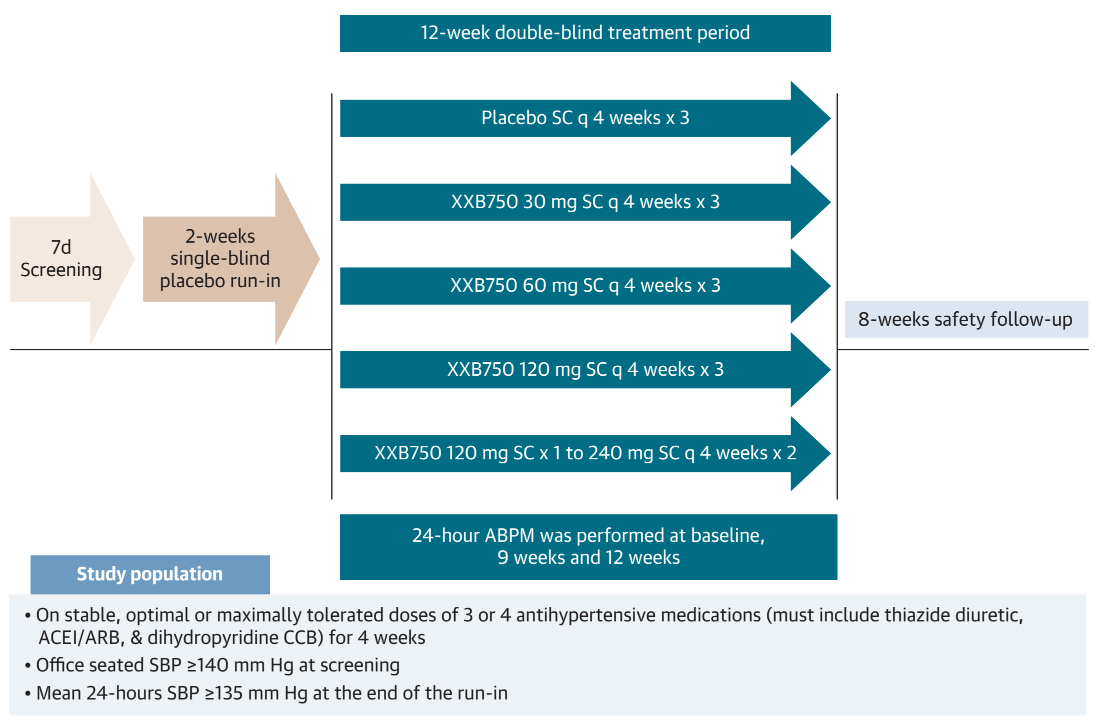
METHODS

This was a 20-week international, multicenter, randomized, double-blind, placebo-controlled, parallel-group, dose-finding phase 2 study comparing 4 doses of XXB750 vs placebo in patients with resistant hypertension. Prespecified primary and key secondary study endpoints were obtained after 12 weeks of double-blind therapy (**Figure 1**). After week 12, an 8-week period of safety assessment with no further administration of the study drug was conducted.

This study was conducted in accordance with the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use for Good Clinical Practice, which has its origin from the Declaration of Helsinki. The study protocol and all amendments were approved by the independent ethics committee or institutional review board of each participating center. All patients signed informed consent before the start of any study-related procedures. The study is registered at ClinicalTrials.gov (NCT05562934). The trial protocol and the statistical analysis plan are available in the Supplemental Appendix.

STUDY POPULATION. Eligible patients were men or women aged ≥18 years with apparent resistant hypertension at screening, defined as mean office seated SBP (office SBP) ≥140 mm Hg despite stable treatment (ie, unchanged for ≥4 weeks) with optimal or maximally tolerated doses of 3 or 4 guideline-directed antihypertensive drugs of different classes, including a diuretic. The main exclusion criteria were history of secondary hypertension, estimated glomerular filtration rate <30 mL/min/1.73 m^2 (CKD-EPI [Chronic Kidney Disease Epidemiology Collaboration] equation), and severe hypertension. Patients were also excluded if they had received treatment with a mineralocorticoid receptor antagonist or sacubitril/valsartan within the 4 weeks before screening. Detailed inclusion and exclusion criteria are presented in the Supplemental Appendix.

FIGURE 1 Study Design



This was a randomized, double-blind, placebo-controlled study of 4 doses of XXB750 and placebo administered for 12 weeks in patients with resistant hypertension. The drug was dosed with subcutaneous (SC) injection at 4-week intervals. The primary endpoint was the change from baseline in 24-hour ambulatory systolic blood pressure (SBP) after 12 weeks of double-blind therapy. ACEI/ARB = angiotensin-converting enzyme inhibitor/angiotensin II receptor blocker; CCB = calcium-channel blocker.

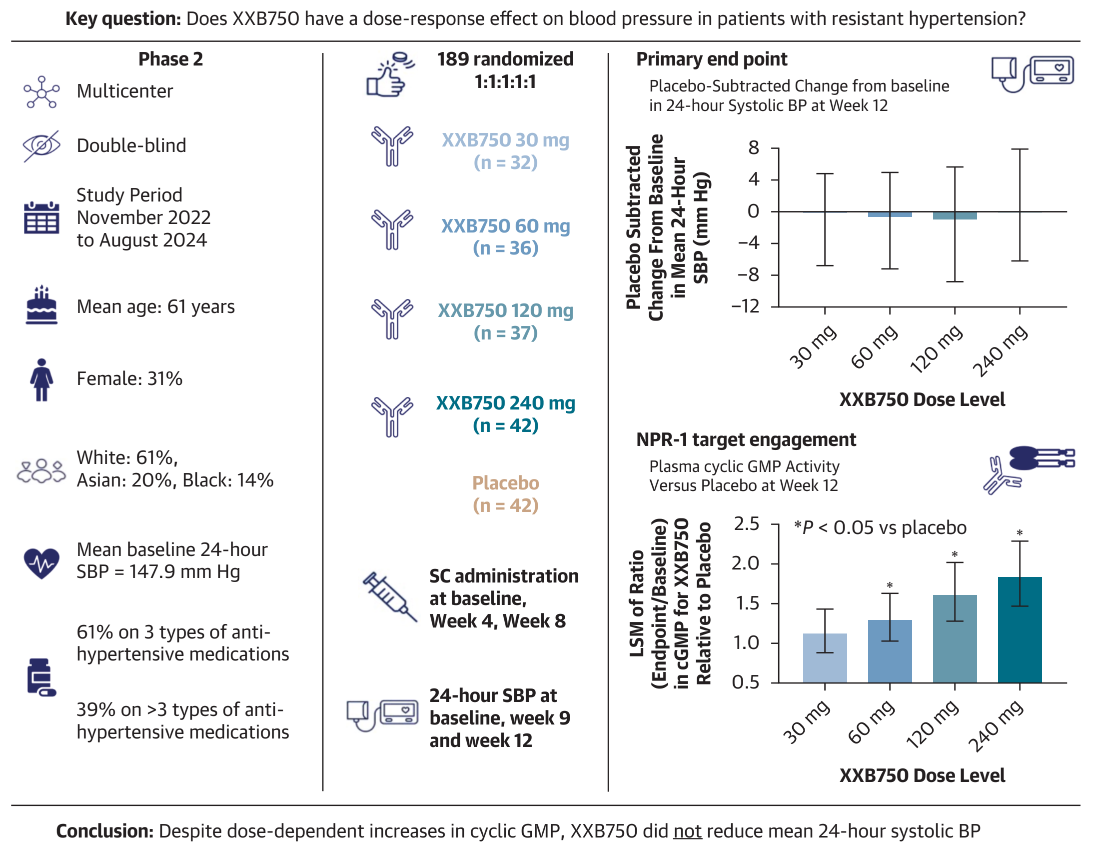
STUDY TREATMENTS. After a 2-week, single-blind, placebo run-in period on background antihypertensive medications, patients with true resistant hypertension (a 24-hour ambulatory SBP ≥135 mm Hg), 24-hour ambulatory SBP/diastolic BP (DBP) ≤170/105 mm Hg, and office SBP/DBP <180/110 mm Hg were randomized (1:1:1:1:1 ratio) to receive monthly SC injection of one of the four XXB750 doses or placebo at baseline, week 4, and week 8, in addition to their daily background antihypertensive therapy. The selected monthly SC doses of XXB750 were 30 mg, 60 mg, 120 mg, or 240 mg (after an initial dose of 120 mg) (Figure 1).

Patients could be randomized to the 240 mg treatment arm only after safety clearance of the lower doses following review by the data monitoring committee. Primary and secondary endpoints were ascertained after 12 weeks of double-blind treatment (Central Illustration, Figure 1). No changes to background antihypertensive therapy were permitted during the 12-week treatment period, except for safety reasons. After the 12-week double-blind randomized treatment, patients entered an 8-week safety follow-up assessment. During this period, patients could receive additional types of antihypertensive therapies at the investigator's discretion, with the consideration that their BP might still be influenced by the study treatment.

STUDY PROCEDURES. From randomization to 12 weeks, patients were seen at monthly morning outpatient visits before intake of their morning background antihypertensive medications and injections of study drug; they underwent safety, office BP, and heart rate measurements, as well as laboratory assessments. A telehealth visit was also conducted 3 days after the first dose and an in-person clinic visit 2 weeks after the first dose of study drug. Adverse events (AEs) and concomitant medications were recorded at each visit, and home BP data were reviewed.

BP MEASUREMENTS. Trained observers measured office BP in triplicate after 5 minutes of rest in the seated position during each study visit. Study

CENTRAL ILLUSTRATION Natriuretic Peptide Receptor 1 Agonist XXB750 for Resistant Hypertension: A Randomized Phase 2 Trial



White WB, et al. JACC. 2026;■(■):■-■.

The left panel presents aspects of the study design and patient characteristics. The middle panel shows the randomization scheme and sample size in each treatment group and notes that the primary endpoint was the change from baseline in the 24-hour systolic blood pressure (SBP). The right panel shows study results including the dose-related increase in the ratio in cyclic guanosine monophosphate (GMP) for XXB750 vs placebo and the placebo-subtracted changes from baseline in 24-hour SBP. NPR-1 = natriuretic peptide receptor-1.

participants also were trained to measure their BP at home after at least 5 minutes of rest in the sitting position in the morning and the evening during 7 consecutive days before each outpatient visit. The same validated digital BP monitor (Microlife, WatchBP Office 2G; Microlife AG) was provided to sites for both the office and home BP measurements. Measurement of 24-hour ambulatory BP (Oscar 2 ambulatory BP recorder; SunTech Medical, Inc) was performed at baseline, 9 weeks, and 12 weeks.

All ambulatory BP data were transferred to a core laboratory, which was masked to treatment assignment, for data cleaning and calculation of the mean 24-hour, daytime, and nighttime values for BP and heart rate. Detailed BP measurement methods are presented in the Supplemental Appendix.

BIOMARKER ASSESSMENTS. Plasma and urine cGMP, plasma renin and serum aldosterone concentrations, and random spot urinary sodium concentrations were measured at baseline and 4, 8, and 12 weeks using validated assays (Supplemental Appendix).

Detection of background antihypertensive medications in serum at baseline and week 12 was assessed by using liquid chromatography tandem mass spectrometry by an independent core laboratory (SGS France SAS) who was masked to the treatment assignment. Full adherence to medications was defined as the presence of 3 prescribed drug classes (angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers, calcium-channel blockers, and diuretics) in the sample.

For pharmacokinetic assessments, blood samples were collected at various time points: 2 to 4 hours postdose at baseline, before the administration of the study drug at week 4, before the administration of the study drug, and 2 to 4 hours postdose at week 8, and at any time during the visits at week 9, week 12, week 16, and week 20. The concentration of XXB750 was measured by using a validated (liquid chromatography tandem mass spectrometry) method (lower limit of quantification of 50 ng/mL).

For immunogenicity assessments, anti-drug antibody (ADA) assays were performed for all patients at baseline, week 4, week 8, and at any time during the visit at weeks 12 and 20 using a validated ligand-binding assay (enzyme-linked immunosorbent assay; Meso Scale Discovery and Arbor Assay). Participants randomized to the placebo group did not undergo testing for ADAs and were assumed to remain ADA negative throughout the study.

STATISTICAL ANALYSES. Prespecified study endpoints. The primary endpoint was to assess the dose-response relationship of XXB750 in reducing the mean 24-hour SBP from baseline to week 12 compared with placebo. The focus of the analyses was change from baseline in 24-hour SBP in each of the 4 treatment groups vs placebo. Secondary endpoints included: 1) evaluation of the effect of the highest dose of XXB750 (240 mg) vs placebo on mean 24-hour SBP from baseline to week 12 and average of mean 24-hour SBP measured at weeks 9 and 12; 2) the dose-related proportion of patients achieving ambulatory BP control (defined as 24-hour SBP <130 mm Hg and diastolic BP [DBP] <80 mm Hg) across the four XXB750 dose groups compared with placebo at week 12; and 3) assessment of safety and tolerability.

Other prespecified assessments included: 1) the proportion of patients achieving office BP control (defined as SBP <130 mm Hg and DBP <80 mm Hg) for XXB750 doses compared with placebo at week 12; 2) the dose-response relationship of XXB750 in reducing 24-hour DBP, office SBP, and office DBP; 3) changes from baseline in biomarkers, including plasma cGMP, urine cGMP/creatinine ratio, serum aldosterone, and plasma renin concentrations; 4) adherence to background antihypertensive medications at baseline and 12 weeks; and 5) the occurrence of ADAs and their impact on the pharmacokinetic variables of XXB750.

Prespecified major AEs assessed in the intention-to-treat population were tabulated according to treatment assignment. An independent data monitoring committee reviewed study data at regular intervals for all enrolled participants; data monitoring committee membership is presented in the Supplemental Appendix.

Model averaging to obtain dose response. A parametric bootstrap-based model averaging approach was used to estimate the dose-response relationship of XXB750. Using this approach, multiple sets of dose-response data and target dose estimates were generated. From these, the mean dose-response values for each XXB750 dose group, as well as the mean differences compared with placebo, were calculated. The 95% CIs for these estimates were derived from the empirical quantiles, specifically the median, 2.5th, and 97.5th percentiles of the bootstrap distributions.

Dose-response results, both absolute changes and placebo-adjusted changes, along with their corresponding 95% CIs, were derived by using a model-averaging approach applied across all candidate models. For comparison, estimates from an analysis of covariance model were also generated by using imputed data sets. These were combined using Rubin's rule and are presented alongside the model-averaged estimates. A graphical representation of the dose-response curve, based on the model-averaging results, was also prepared to illustrate the shape and trend of the response across dose levels. An analysis of covariance model was run with the observed change from baseline at week 12 as response, treatment, number of background antihypertensive medications (3 or >3), center, and baseline 24-hour SBP as covariates and center-by-treatment as the interaction. The P value for the center-by-treatment interaction was not significant ($P = 0.446$).

ANALYTICAL METHODS. The primary endpoint was analyzed by using an optimally weighted contrast test following the MCP-Mod methodology, testing the hypothesis of a flat dose-response curve in the full analysis set (FAS) population. The MCP-Mod methodology is a statistical framework that combines the strengths of multiple comparison procedures (MCP) with model-based dose-response estimation (Mod). Five candidate dose-response models (eg, linear, E_{max}, sigmoidal, beta-model) were tested to identify a statistically significant 24-hour ambulatory BP dose-response relationship at week 12 while controlling the family-wise error rate. These models are described in detail in the Supplemental Appendix. The randomization of the study was stratified according to region (which was included in the prespecified primary analysis), MCP-Mod, and prespecified supplemental analysis of mixed models for repeated measures as a fixed effect factor. The estimated treatment effect was adjusted for region in the prespecified analyses.

Multiplicity adjustment for comparisons were conducted as follows. A candidate model set defined on the range of the expected mean response in each dose group was used to generate a set of optimal contrasts vectors to test the null hypothesis of a flat dose-response curve. The optimal contrasts derived from the candidate model set was applied to the combined estimated dose means and covariance matrix to obtain t statistics for each candidate model and the common critical value $C_{0.025}$. $C_{0.025}$ was the common critical value derived from the reference multivariate t-distribution with the 5×5 correlation matrix induced by testing the candidate dose-response models. The null hypothesis was rejected, and the statistical significance of dose-response data in mean 24-hour SBP reduction was to be established if the maximum (t1, t2,..., t5) $\geq C_{0.025}$.

Sample size estimation and power calculations. A total of 170 patients were planned to be randomized, allowing for an anticipated 10% loss to follow-up, with equal allocation (1:1:1:1:1) to placebo and each of the four XXB750 dose groups. Assuming a common SD of 12 mm Hg for the change in mean 24-hour SBP from baseline, and applying a one-sided significance level of 2.5% with adjustment for multiple comparisons using the MCP-Mod approach, a sample size of 150 participants (30 per group) was estimated to provide at least 92% power to detect a dose-response signal. This calculation assumed a true maximum treatment effect of a 10-mm Hg reduction in mean 24-hour SBP for XXB750 compared with placebo.

For secondary endpoints, the same assumptions yielded approximately 89% power to detect a 10-mm Hg difference in mean 24-hour SBP between placebo and the highest XXB750 dose group (240 mg) at individual time points (weeks 9 and 12). Furthermore, the study was powered at 76% to detect differences in ambulatory BP control rates ranging from 10% in the placebo group to 45% across the XXB750-treated groups, under the same sample size and MCP-Mod adjustment framework.[10,11]

The analyses were performed by using SAS version 9.4 (SAS Institute, Inc). A 1-sided *P* value <0.025 or a 2-sided *P* value <0.05 were considered significant.

RESULTS

PATIENT DISPOSITION. The study was conducted between November 2022 and August 2024. The patient disposition flowchart is presented in Supplemental Figure 1. Of the 629 screened patients, 306 (48.7%) did not meet the run-in entry criteria, with the most common reason being failure to meet the required BP criteria at screening (29.6%). Of the 323 remaining patients, 190 completed the run-in period and were randomized to treatment; one patient inadvertently was randomized without completing the run-in period. The randomized population included 191 patients, including 2 patients who were mis-randomized (ie, randomized by error and never received study medication); the FAS population comprised 189 patients, with 180 (94.2%) patients completing the study. Fourteen (7.4%) patients discontinued the study treatment prematurely, 6 of whom discontinued due to AEs.

BASELINE DEMOGRAPHIC CHARACTERISTICS. The baseline demographic and disease characteristics of the patients are presented in **Table 1** and were balanced among the randomized groups. Mean age ranged from 58.5 to 63.4 years. Among the 189 randomized patients, 30.7% were women, 14.3% were Black/African American, 42.9% had diabetes, and 14.8% had an estimated glomerular filtration rate <60 mL/min/1.73 m^2. The mean body mass index ranged from 31.1 to 33.6 kg/m^2. All patients had true resistant hypertension confirmed by ambulatory BP monitoring despite being treated with 3 or 4 background antihypertensive medications (61.4% and 38.6%, respectively); office SBP/DBP was 152.6/87.6 mm Hg, and 24-hour SBP/DBP was 147.9/81.0 mm Hg. The percentage of patients taking diuretics, renin-angiotensin system blockers (angiotensin-converting enzyme inhibitors and angiotensin II receptor blockers), beta-blockers, and calcium-channel blockers is shown in Supplemental Table 1.

PRIMARY EFFICACY RESULTS. There was no significant XXB750 dose-response relationship observed for the change from baseline in mean 24-hour SBP at week 12 (**Figure 2**, **Table 2**). The BP-lowering dose response to XXB750 did not significantly differ from that observed with placebo (XXB750 dose groups: −5.4 to −6.7 mm Hg vs placebo: −5.8 mm Hg) (Supplemental Figure 2).

SECONDARY EFFICACY RESULTS. XXB750 240 mg dose vs placebo. The 24-hour ambulatory SBP change from baseline for the XXB750 240 mg dose and placebo was −4.6 mm Hg and −5.5 mm Hg at week 9 (1 week postdose) and −4.6 mm Hg and −6.0 mm Hg at week 12 (4 weeks postdose), respectively (*P* = 0.797 and 0.696). The averaged 24-hour ambulatory SBP change from baseline for the 2 time points (ie, weeks 9 and 12) was −4.6 mm Hg for the XXB750 240 mg dose and −5.8 mm Hg for placebo (*P* = 0.711) (Supplemental Figure 3).

TABLE 1 Characteristics of the Study Participants (FAS) at Baseline

	Placebo (n = 42)	XXB750 30 mg (n = 32)	XXB750 60 mg (n = 36)	XXB750 120 mg (n = 37)	XXB750 240 mg (n = 42)
Age, y	60.1 ± 10.9	58.5 ± 11.6	63.4 ± 10.2	62.4 ± 11.6	60.6 ± 10.1
Female	13 (31.0)	8 (25.0)	6 (16.7)	18 (48.7)	13 (31.0)
Race					
White	27 (64.3)	16 (50.0)	24 (66.7)	24 (64.9)	25 (59.5)
Asian	8 (19.1)	5 (15.6)	6 (16.7)	8 (21.6)	10 (23.8)
Black or African American	6 (14.3)	7 (21.9)	4 (11.1)	4 (10.8)	6 (14.3)
Other	1 (2.4)	4 (12.5)	2 (5.6)	1 (2.7)	1 (2.4)
Ethnicity					
Hispanic or Latino	5 (11.9)	3 (9.4)	6 (16.7)	3 (8.1)	6 (14.3)
BMI, kg/m^2	33.6 ± 5.8	32.0 ± 4.8	31.9 ± 6.1	32.5 ± 6.3	31.1 ± 4.6
No. of background antihypertensive medications at randomization[a]					
3	28 (66.7)	24 (75.0)	22 (61.1)	21 (56.8)	21 (50.0)
>3	14 (33.3)	8 (25.0)	14 (38.9)	16 (43.2)	21 (50.0)
Office BP, mm Hg					
Systolic	150.4 ±10.8	150.9 ± 13.7	156.5 ± 12.7	152.2 ± 13.3	153.0 ± 14.7
Diastolic	88.1 ± 9.5	89.9 ± 12.0	88.6 ± 9.8	84.3 ± 10.5	87.1 ± 13.0
Mean 24-hour ambulatory BP, mm Hg					
Systolic	148.6 ± 12.7	147.9 ± 8.1	145.9 ± 13.0	149.2 ± 9.4	147.6 ± 8.5
Diastolic	82.2 ± 10.3	84.6 ± 9.9	78.9 ± 9.2	80.3 ± 11.8	79.6 ± 9.3
eGFR, mL/min/1.73 m^2	86.5 ± 15.7	81.8 ± 22.1	79.3 ± 21.4	80.0 ± 23.6	86.2 ± 19.3
eGFR <60 mL/min/1.73 m^2	3 (7.1)	5 (15.6)	9 (25.0)	7 (18.9)	4 (9.5)
Diabetes mellitus	13 (31.0)	14 (43.8)	18 (50.0)	15 (40.5)	21 (50.0)
History of coronary heart disease	5 (11.9)	0 (0)	5 (13.9)	2 (5.4.1)	4 (9.5)
Hyperlipidemia	7 (16.7)	8 (25.0)	8 (22.2)	9 (24.3)	14 (33.3)

Values are mean ± SD or n (%). [a]Details on background antihypertensive treatment are provided in Supplemental Table 1.
BP = blood pressure; BMI = body mass index; eGFR = estimated glomerular filtration rate; FAS = full analysis set; n = number of patients with respective categories.

Proportion of patients achieving BP control. At week 12, the proportion of patients achieving BP control with XXB750 240 mg was 25.9%. However, no significant dose-response relationship was observed for any of the tested candidate models (30 mg: 15.0%; 60 mg: 17.7%; 120 mg: 22.0%; placebo: 13.2%).

EXPLORATORY ENDPOINTS. Office BP. There were numerically greater changes from baseline in office SBP across the XXB750 dose groups compared with placebo in the first 4 weeks after the first dose of study medication. However, by week 12, there was no significant dose-response effect on mean office SBP observed across any of the candidate models (Figure 3, Table 2, Supplemental Table 2). However, a significant dose-response signal was observed for the proportion of patients achieving office BP control at week 12 (placebo: 2.4%; 30 mg: 12.5%; 60 mg: 16.7%; 120 mg: 24.3%; 240 mg: 7.1%; two-sided $P = 0.0214$).

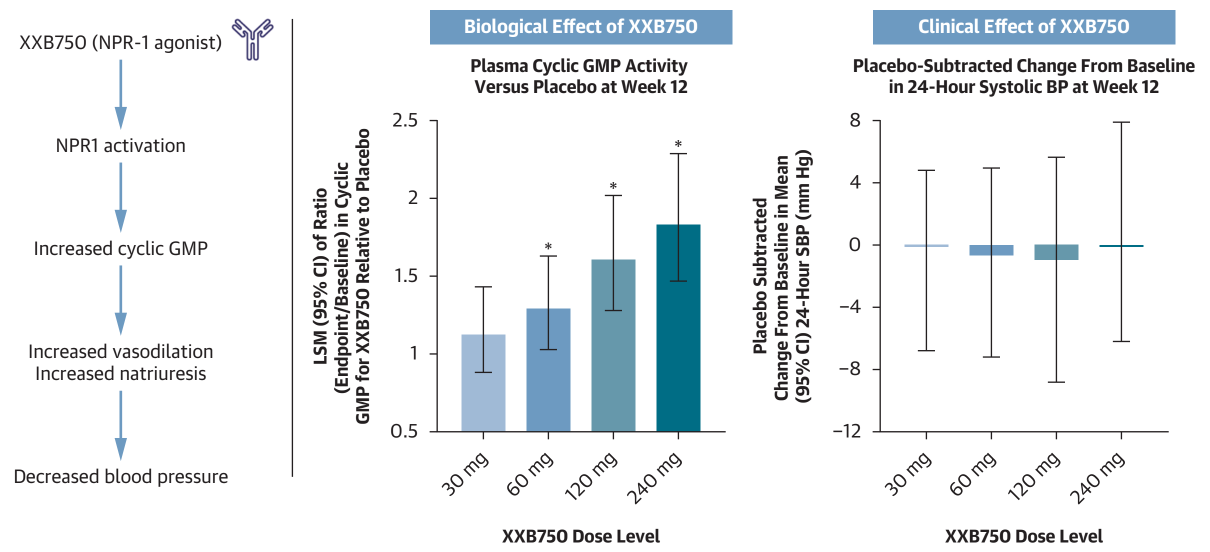
Home BP. There was no dose-response relationship observed for changes from baseline in home SBP or DBP across the XXB750 doses tested (Supplemental Tables 3 and 4).

Heart rate. There were no clinically meaningful changes in the office heart rate throughout the 12 weeks of study with XXB750 (Supplemental Table 5).

BIOMARKER DATA. Plasma cGMP. The plasma cGMP concentrations dose-dependently increased from baseline to week 12 in the XXB750 groups relative to placebo (Figure 2, Table 3). A significant dose-response signal for changes in both plasma cGMP and urine cGMP/creatinine ratios were also observed across all candidate models.

Plasma renin and serum aldosterone. Plasma renin concentrations (geometric mean ratios) increased across all XXB750 dose groups compared with placebo from baseline to week 12, although only the results for the 120 mg group reached marginal statistical significance (Table 3). Change in serum aldosterone at week 12 was also marginally statistically significant compared with placebo in the 120 mg

FIGURE 2 Primary and Secondary Results of the Trial



Expected biological effect of natriuretic peptide receptor-1 (NPR-1) agonist activity is shown in the left panel. The study results showed a robust dose-related effect of XXB750 on cyclic guanosine monophosphate (GMP) activity (middle panel) but a flat dose-response on ambulatory systolic blood pressure (BP) by treatment dose (right panel). All comparisons are between active treatment groups and placebo. **P* < 0.05 vs placebo. LSM = least square means.

group. Spot urinary sodium excretion was comparable to placebo across the XXBB750 doses.

Body weight and kidney function. There were no clinically important changes from baseline in body weight in any of the 5 treatment groups (Supplemental Table 6). In addition, no clinically relevant changes from baseline in the estimated glomerular filtration rate, serum potassium, or urinary albumin-creatinine levels were observed in any of the treatment groups (Supplemental Table 7).

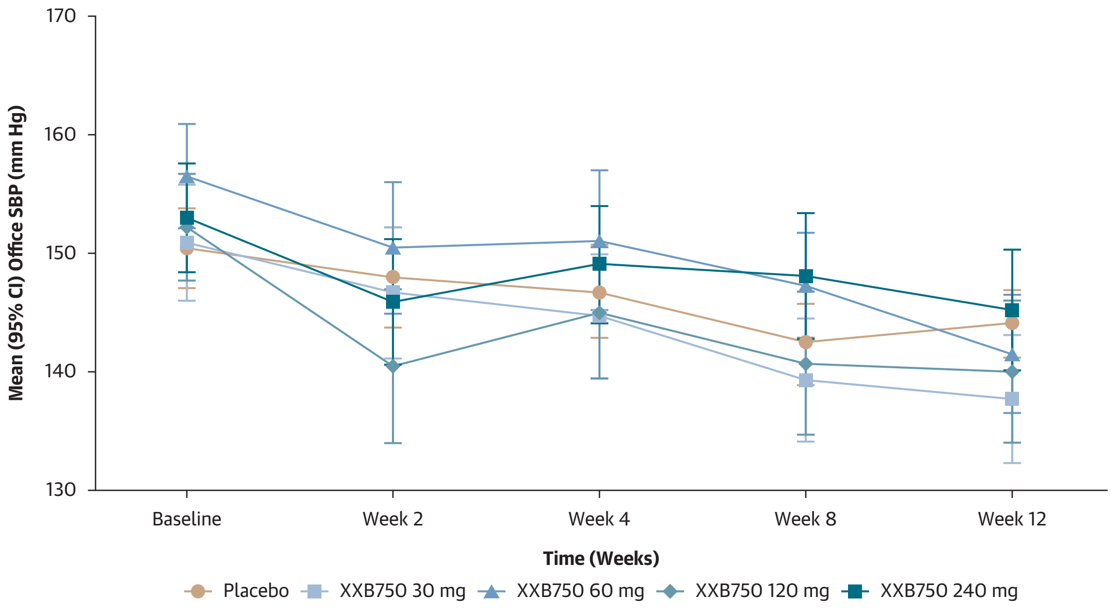
TABLE 2 Changes From Baseline in 24-Hour Ambulatory, Office, and Home SBP at 12 Weeks

Type of Blood Pressure Measurement (mm Hg)	Placebo (n = 42)	XXB750 30 mg (n = 32)	XXB750 60 mg (n = 36)	XXB750 120 mg (n = 37)	XXB750 240 mg (n = 42)
Baseline 24-hour SBP	148.6 ± 12.7	147.9 ± 8.1	145.9 ± 13.0	149.2 ± 9.4	147.6 ± 8.5
Between-group changes from baseline (95% CI) in 24-hour SBP	-	−0.1 (−6.8 to 4.8)	−0.7 (−7.2 to 4.9)	−1.0 (−8.8 to 5.6)	−0.1 (−6.2 to 7.9)
Within-group (95% CI) changes from baseline in 24-hour SBP	−5.8 (−10.7 to −1.1)	−6.4 (−10.2 to −2.9)	−6.6 (−10.4 to −3.5)	−6.7 (−12.6 to −3.2)	−5.4 (−9.6 to −0.4)
Baseline office SBP	150.4 ± 10.8	150.9 ± 13.7	156.5 ± 12.7	152.2 ± 13.3	153.0 ± 14.7
Between-group changes from baseline (95% CI) in office SBP	-	−4.9 (−10.7 to 0.9)	−5.1 (−10.0 to 1.7)	−3.8 (−9.5 to 3.3)	−0.3 (−6.2 to 7.8)
Within-group (95% CI) changes from baseline in Office SBP	−7.5 (−13.1 to −3.8)	−12.2 (−17.1 to −8.3)	−12.4 (−15.9 to −9.0)	−11.3 (−15.6 to −8.0)	−8.1 (−12.5 to −3.9)
Baseline home SBP	146.5 ± 12.6	145.2 ± 14.2	148.4 ± 15.9	149.5 ± 8.3	150.1 ± 11.8
Between-group changes from baseline (95% CI) home SBP	-	−0.8 (−5.9 to 3.1)	−2.9 (−8.4 to 3.1)	−5.8 (−11.6 to 3.1)	−0.5 (−6.6 to 7.1)
Within-group change (95% CI) from baseline in Home SBP	−4.8 (−10.2 to −1.4)	−6.3 (−10.1 to −2.8)	−8.0 (−11.8 to −4.7)	−10.3 (−15.3 to −5.5)	−5.8 (−10.3 to −1.3)

Baseline values are mean ± SD in millimeters of mercury; changes from baseline are presented as mean and 95% CIs in millimeters of mercury using the multiple comparison procedure-modeling methodology. Ns represent the full analysis set.

SBP = systolic blood pressure.

FIGURE 3 Time Course of Office SBP According to Treatment Group During the Study Period



XXB750 was administered at baseline, 4 weeks, and 8 weeks by subcutaneous injection. The bars are 95% CIs. None of the between-group differences is statistically significant. SBP = systolic blood pressure.

BACKGROUND ANTIHYPERTENSIVE THERAPY DETECTION. There were 88 and 85 patients with adequate blood samples at baseline and week 12, respectively, for adherence testing. Of these, 57% were fully adherent to 3 background antihypertensive medication classes at baseline and 53% at week 12 (Table 4). In addition, at both baseline and week 12, the proportions of study participants who were fully adherent to their background therapy were comparable by treatment group.

Using multiple imputation for missing data, the mean 24-hour SBP reduction increased modestly as the XXB750 dose increased among those participants who had 3 background medications detected at both baseline and week 12 (Table 5). However, the increase in reduction was not statistically significant nor clinically relevant. The monotonic increasing dose-response relationship in mean 24-hour SBP reduction was not observed among those without adherence at either baseline or at week 12.

ADA AND PHARMACOKINETIC PARAMETERS OF XXB750. Most patients receiving XXB750 developed ADAs at some point in the study but at low titers. Low titers were derived from observed titer ranges and their distribution shapes. Although titers ranged from thousands to single digits, most titers in ADA-positive subjects were in single to low double digits. Of the 124 study participants with known ADA status at week 12, 60.5% were ADA positive, and 39.5% were ADA negative. There were no consistent differences for the changes from baseline in the 24-hour SBP at week 12 in the active drug treatment groups for those were ADA positive vs those who did not develop ADAs (Table 6).

Pharmacokinetic analyses showed a dose-proportional increase in XXB750 exposure according to dose (Supplemental Figure 4). There was interference in XXB750 exposure with the development of ADA titers, particularly when ≥1:32, but this occurred in the minority of study participants in the trial (Supplemental Figure 4, Supplemental Table 8).

SAFETY AND TOLERABILITY. No deaths were reported during the study. No serious AEs (SAEs) occurred in the placebo group. Seven (3.7%) patients reported at least one SAE in the XXB750 groups (Table 7). The most frequently reported SAE during the randomized treatment period was pneumonia, occurring in 3 (1.6%) patients with multiple comorbidities receiving XXB750 (Figure 4, Table 7). The onset of pneumonia among these patients occurred over a relatively broad timeframe, ranging from 3 weeks to 4 months after the initiation of treatment. All affected individuals exhibited one or more pre-existing risk factors known to predispose to

TABLE 3 Comparative Changes in Prespecified Biomarkers for XXB750 Doses vs Placebo After 12 Weeks of Therapy

Treatment Group	n/N	Baseline Geometric Mean (95% CI)	Least Square Means Ratio of Change in XXB750 vs Placebo (95% CI)	Nominal 2-Sided *P* Value
Plasma cGMP, nmol/L				
Placebo	37/42	5.7 (4.3-7.7)	-	-
XXB750 30 mg	26/32	5.3 (4.5-6.2)	1.1 (0.9-1.4)	0.3390
XXB750 60 mg	31/36	5.6 (4.7-6.5)	1.3 (1.0-1.6)	0.0288
XXB750 120 mg	33/37	5.2 (4.3-6.1)	1.6 (1.3-2.0)	<0.0001
XXB750 240 mg	38/42	5.1 (4.5-5.9)	1.8 (1.5-2.3)	<0.0001
Urinary cGMP/creatinine ratio, nmol/nmol				
Placebo	36/42	63.4 (53.9-74.6)	-	-
XXB750 30 mg	26/32	52.9 (42.8-65.3)	1.1 (0.9-1.4)	0.5510
XXB750 60 mg	31/36	57.0 (46.8-69.3)	1.1 (0.9-1.4)	0.3436
XXB750 120 mg	32/37	61.5 (51.5-73.5)	1.4 (1.1-1.7)	0.0032
XXB750 240 mg	38/42	63.2 (53.0-75.3)	1.6 (1.3-1.9)	<0.0001
Plasma renin concentration, pg/mL				
Placebo	39/42	12.2 (7.8-19.2)	-	-
XXB750 30 mg	25/32	19.6 (11.1-34.5)	1.4 (0.8-2.5)	0.2238
XXB750 60 mg	33/36	17.2 (8.6-34.3)	1.1 (0.7-1.9)	0.6817
XXB750 120 mg	30/37	7.7 (4.5-13.0)	1.7 (1.0-3.0)	0.0513
XXB750 240 mg	35/42	9.5 (5.7-16.0)	1.4 (0.8-2.3)	0.2148
Serum aldosterone, nmol/L				
Placebo	36/42	287.8 (244.3-339.0)	-	-
XXB750 30 mg	23/32	323.2 (260.8-400.6)	1.0 (0.8-1.2)	0.8103
XXB750 60 mg	31/36	277.0 (211.6-362.7)	0.9 (0.8-1.2)	0.5617
XXB750 120 mg	31/37	252.8 (200.0-319.5)	1.2 (1.0-1.5)	0.0572
XXB750 240 mg	34/42	274.0 (218.9-343.0)	1.0 (0.8-1.3)	0.7801
Urinary sodium, mmol/L				
Placebo	37/42	86.9 (73.1-103.4)	-	-
XXB750 30 mg	26/32	65.4 (48.3-88.4)	1.0 (0.7-1.3)	0.8489
XXB750 60 mg	31/36	73.2 (57.0-94.1)	1.1 (0.8-1.4)	0.6897
XXB750 120 mg	33/37	92.0 (74.6-113.4)	0.9 (0.7-1.1)	0.3697
XXB750 240 mg	38/42	78.5 (63.0-97.8)	1.0 (0.8-1.3)	0.9483

cGMP = cyclic guanosine monophosphate; LSM = least squares mean; n = number of patients with a non-missing value at both baseline and this corresponding visit; N = full analysis set patients.

pneumonia, including the use of tumor necrosis factor inhibitors, chronic cigarette smoking, chronic pulmonary disease, and diabetes. All patients fully recovered with standard antibiotic therapy. All other SAEs were reported in single patients (1 in the 30 mg group and 3 in the 240 mg group). All SAEs resolved, and none was considered related to the study treatment. There were 78 (41.3 %) patients who experienced at least one AE, and 20 (10.6%) patients had AEs suspected to be related to the study treatment. There were no dose-related trends or differences in the incidence rates of reported AEs.

DISCUSSION

The principal finding of this study was that the human monoclonal antibody NPR-1 agonist XXB750 had neutral effects on ambulatory, office, and home BP in patients with resistant hypertension, despite showing a clear dose-response relationship with respect to engaging the target of pharmacologic action (ie, NPR-1) as indicated by increases in cGMP levels. These findings were unexpected considering the results from an earlier phase 1 study with XXB750 that found a significant dose-related reduction from baseline in BP after single doses of 30 to 600 mg.[6] In that study, XXB750 was evaluated in 73 subjects with normotension to high normal office BP not receiving antihypertensive therapies and followed up for 3 months. At 28 days postdosing, there were persistent reductions in 24-hour ambulatory SBP by 8 to 16 mm Hg after administration of the intermediate and higher doses (≥120 mg). In addition, the target engagement biomarker cGMP was increased for up to 3 months after a single dose, providing biological evidence that XXB750 activates the NPR-1 receptor.

TABLE 4 Background Antihypertensive Medication According to Number of Detectable Medications per Treatment Groups

Visit	Background Medication Classes Detectable	Placebo (n = 42) n/m (%)	XXB750 30 mg (n = 32) n/m (%)	XXB750 60 mg (n = 36) n/m (%)	XXB750 120 mg (n = 37) n/m (%)	XXB750 240 mg (n = 42) n/m (%)	Total (N = 189) n/m (%)
Baseline							
	No background medication class detectable	1/22 (4.5)	1/16 (6.3)	0/18 (0.0)	2/13 (15.4)	2/19 (10.5)	6/88 (6.8)
	1 background medication class detectable	0/22 (0.0)	2/16 (12.5)	2/18 (11.1)	1/13 (8.3)	1/19 (5.3)	6/88 (6.8)
	2 background medication classes detectable	7/22 (31.8)	6/16 (37.5)	4/18 (22.2)	3/13 (23.1)	6/19 (31.6)	26/88 (29.6)
	3 background medication classes detectable	14/22 (63.6)	7/16 (43.8)	12/18 (66.7)	7/13 (53.9)	10/19 (52.6)	50/88 (56.8)
Week 12							
	No background medication class detectable	0/22 (0.0)	0/14 (0.0)	0/16 (0.0)	2/15 (13.3)	3/18 (16.7)	5/85 (5.9)
	1 background medication class detectable	0/22 (0.0)	0/14 (0.0)	1/16 (6.3)	1/15 (6.7)	1/18 (5.6)	3/85 (3.5)
	2 background medication classes detectable	9/22 (40.9)	7/14 (50.0)	6/16 (37.5)	5/15 (33.3)	5/18 (27.8)	32/85 (37.7)
	3 background medication classes detectable	13/22 (59.1)	7/14 (50.0)	9/16 (56.3)	7/15 (46.7)	9/18 (50.0)	45/85 (52.9)

Values are number of full-analysis set (FAS) participants with related number of detectable background medication classes at each visit (n)/number of FAS participants who got tested in all 3 classes and considered in the analysis (m) (%): n/m (%). N indicates the number of FAS participants. % = n/m × 100. A participant with multiple detected medications within a background medication class is counted only once.

In contrast to the phase 1 trial of XXB750,[6] the current phase 2 trial used a different multiple-dose study design and involved a much more complex population of patients with hypertension whose disease was uncontrolled despite guideline-based therapy that included a diuretic, renin-angiotensin system blocking agent, dihydropyridine calcium-channel blocker, and a beta-blocker if needed. There was an unexpectedly large decrease in 24-hour ambulatory SBP in the placebo arm of the current study of approximately 6 mm Hg. Compared with office BP measurements, BP values obtained through ambulatory monitors are typically less likely to decline with placebo because observer bias is diminished.[12] However, in more recent trials of resistant hypertension, 3 to 7 mm Hg reductions in 24-hour ambulatory SBP with placebo are becoming commonly observed.[13,14] Hence, we evaluated several behavioral, clinical, and pharmacologic factors to determine potential explanations for the neutral BP effect despite significant engagement of the pharmacologic target cGMP in this first phase 2 trial of XXB750 in patients with resistant hypertension.

The large between-subject variability in ambulatory, office, and home SBP measurements and the substantial reduction in ambulatory and office SBP observed with placebo contributed to a lack of a detectable dose-response pattern in the study. One hypothesis for such findings is that study patients' behavior changed during the trial, the so-called Hawthorne effect, a type of human behavioral reactivity in which study participants modify an aspect of their behavior (eg, background medication

TABLE 5 Subgroup Analysis for Changes in 24-Hour SBP Based on Medication Adherence (Multiple Imputation Model)

No. of Detected Background Medications	Treatment: Treatment Group	Within Treatment: LSM (SE) of Change From Baseline	LSM of Difference (XXB750 vs Placebo): (95% CI)	Treatment-by-Subgroup Interaction: *P* Value
3 at both baseline and week 12				
	Placebo	−5.16 (3.61)		0.9064
	XXB750 30 mg	−5.45 (5.85)	−0.29 (−13.53 to 12.95)	
	XXB750 60 mg	−6.84 (4.53)	−1.68 (−12.88 to 9.52)	
	XXB750 120 mg	−6.95 (6.22)	−1.78 (−15.83 to 12.26)	
	XXB750 240 mg	−8.99 (5.26)	−3.83 (−16.21 to 8.55)	
<3 at baseline or week 12				
	Placebo	−6.78 (3.64)		
	XXB750 30 mg	−6.87 (3.58)	−0.10 (−10.08 to 9.88)	
	XXB750 60 mg	−6.02 (3.93)	0.76 (−9.73 to 11.24)	
	XXB750 120 mg	−7.90 (3.57)	−1.13 (−11.02 to 8.76)	
	XXB750 240 mg	−2.78 (3.26)	3.99 (−5.50 to 13.48)	

LSM = least squares mean; SBP = systolic blood pressure.

TABLE 6 Effect of ADA Status on Dose Response of 24-Hour SBP at Week 12

	Changes in 24-Hour SBP ADA All Participants			Changes in 24-Hour SBP ADA-Negative Participants		
Treatment	**n**	**Model-Based Estimate (95% CI)**	**ANCOVA Estimated (95% CI)**	**n**	**Model-Based Estimate (95% CI)**	**ANCOVA Estimated (95% CI)**
Placebo	40	−5.8 (−10.7 to −1.1)	−5.9 (−10.7 to −1.2)	40	−5.5 (−11.4 to −0.6)	−5.8 (−10.9 to −0.7)
30 mg	21	−6.4 (−10.2 to −2.9)	−6.4 (−12.1 to −0.7)	4	−6.5 (−14.9 to −1.6)	−10.3 (−23.3 to 2.8)
60 mg	19	−6.6 (−10.4 to −3.5)	−6.6 (−11.9 to −1.1)	14	−7.1 (−13.5 to −2.4)	−4.8 (−13.1 to 3.6)
120 mg	20	−6.7 (−12.6 to −3.2.4)	−7.7 (−12.9 to −2.5)	12	−7.9 (−19.9. to −2.1)	−11.7 (−20.7 to −2.7)
240 mg	15	−5.4 (−.6 to −0.4)	−4.7 (−9.6 to 0.2)	21	−5.3 (−11.2 to 2.1)	−4.1 (−10.8 to 2.7)
		MCP-Mod, $P = 0.57$			MCP-Mod, $P = 0.49$	

ADA = anti-drug antibody; ANCOVA = analysis of covariance; MCP-Mod = multiple comparison procedure-modeling; SBP = systolic blood pressure.

adherence) in response to their awareness of being observed.[15] However, in our prespecified serum adherence testing for background antihypertensive drug therapies, while general adherence to a full ≥3-drug background regimen was only about 55%, it was comparable at baseline and at the end of 12 weeks of randomized therapy, as well as in each of the treatment arms. Further analysis did not show a significant difference in the dose-response for those study participants who were adherent vs nonadherent in taking their background antihypertensive therapy.

Another possible cause of reduced efficacy of a monoclonal antibody such as XXB750 is the development of ADAs after multiple dose administrations. ADAs can bind to XXB750 and prevent it from reaching its target or interfere with its mechanism of action, alter its pharmacokinetic parameters, and diminish its therapeutic effect.[16] Most patients in this trial developed ADAs by week 12; however, the titers were low, and, importantly, there was no evidence that those who developed ADAs had a less robust effect on ambulatory BP reduction compared with those who remained ADA negative.

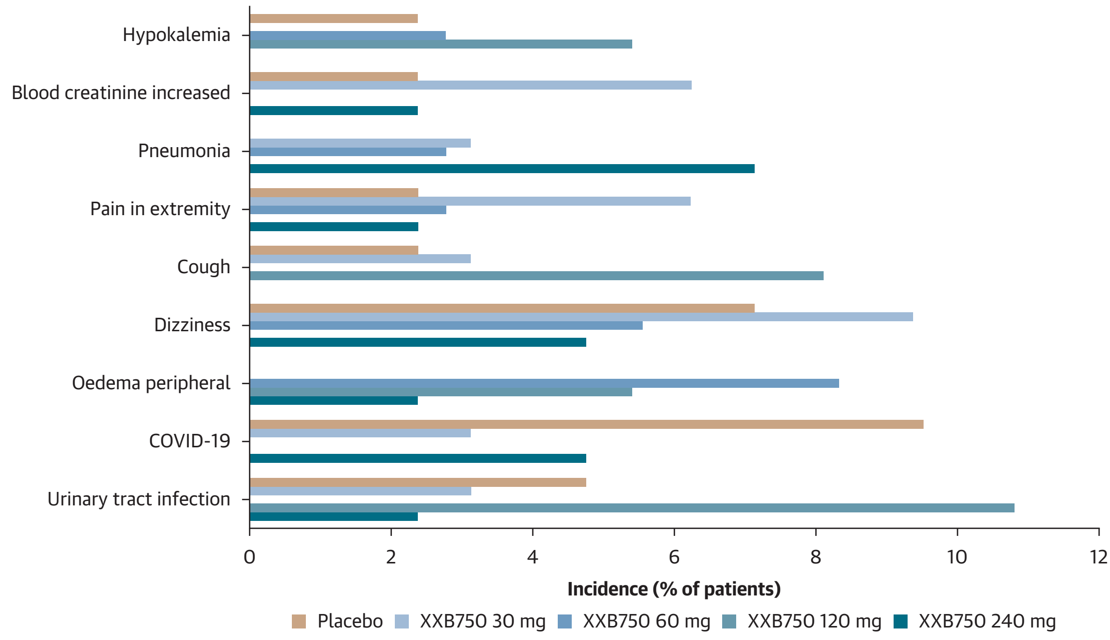
A natriuretic effect of XXB750 was not observed during the 12-week study period, a finding which was similar to the phase 1 study of XXB750 in healthy volunteers,[6] but there was a small increase in plasma renin concentration without any relevant change in serum aldosterone concentrations. Given that XXB750 activates the NPR-1, pharmacologic and physiological actions of vasodilation, natriuresis, and diuresis such as those observed with the natriuretic peptide ANP were expected to occur.[7,17] ANP regulates sodium reabsorption at the apical and basolateral sides in the medullary collecting duct.[18] The epithelial sodium channel for sodium entry is regulated by ANP via NPR-1 receptors located on the medullary collecting duct. There was clear evidence that XXB750 induced the secondary messenger cGMP in a dose-related fashion, which is consistent with the drug influencing NPR-1 receptors and thereby inducing natriuresis. However, the timing of

TABLE 7 AEs During Randomized Double-Blind Treatment Period

	Placebo (n = 42) n (%)	**XXB750 30 mg (n = 32)**	**XXB750 60 mg (n = 36)**	**XXB750 120 mg (n = 37)**	**XXB750 240 mg (n = 42)**	**Total (N = 189)**
Any event	18 (42.9)	17 (53.1)	15 (41.7)	13 (35.1)	15 (35.7)	78 (41.3)
Possibly related to study treatment	3 (7.1)	2 (6.3)	6 (16.7)	2 (5.4)	7 (16.7)	20 (10.6)
Serious AEs, total	0 (0)	2 (6.3)	2 (5.6)	1 (2.7)	2 (4.8)	7 (3.7)
Categories of serious AEs						
Cardiac	0 (0)	0 (0)	1 (2.8)	0 (0)	0 (0)	1 (0.5)
Infections	0 (0)	1 (3.1)	2 (5.6)	0 (0)	2 (4.8)	5 (2.7)
Metabolism and nutritional disorders	0 (0)	0 (0)	0 (0)	1 (2.7)	0 (0)	1 (0.5)
Syncope	0 (0)	1 (3.1)	0 (0)	0 (0)	0 (0)	1 (0.5)
Renal	0 (0)	1 (3.1)	0 (0)	0 (0)	0 (0)	1 (0.5)

Values are number of participants experiencing respective adverse event (AE)/number of participants with serious AE. Study participants with multiple serious AEs within a primary system organ class are counted only once in the total row. System organ classes are presented in alphabetical order; preferred terms are presented in descending order of frequency as reported in Total column. MedDRA version 27.0 was used for the reporting of AEs.

FIGURE 4 **Incidence of Investigator-Reported Adverse Events According to Treatment Group (> 5% in Any Category)**



The most commonly reported adverse events are shown according to treatment assignment over the entire study period. There were no deaths during the trial.

collections of urine for sodium excretion may have played a role in the negative natriuretic findings for XXB750, as these collections were conducted after 8 days of drug administration in the phase 1 study[6] and at 4-week intervals postdose in the current study. These results are consistent with findings for thiazide and loop diuretics, which are known to induce natriuresis for several hours followed by rebounds in sodium retention to maintain homeostasis of sodium balance.[19]

Findings similar to both the phase 1 study[6] and the current XXB750 study were noted for another new NPR-1 agonist, REGN5381, in which 48 healthy subjects were randomized to receive a single intravenous injection of the monoclonal antibody REGN5381 (0.3, 1, 3, 10, 30 or 100 mg) or placebo.[20] The highest REGN5381 dose (100 mg) was associated with a 6 to 9 mm Hg reduction in SBP, a small increase in heart rate, and a reduction in stroke volume throughout a 72-hour period of observation. These hemodynamic changes were associated with a sustained increase in urine cGMP. Notably, as was also seen in preclinical studies, there were no changes in urine flow or urinary sodium output across all dose cohorts. This finding may also be a function of the monoclonal antibody not being filtered to reach the NPR-1.

Finally, preclinical[21] and human subject[7] studies of a modified form of ANP (MANP) have shown increases in urinary flow and urinary sodium excretion for 30 minutes and 4 hours, respectively, after acute administration, while BP-lowering effects were sustained for up to 120 minutes and 12 hours in preclinical models and human subjects. Further information on other NPR-1 agonists in development is presented in Supplemental Table 9. Although some studies (eg, with MANP) are of short duration, it does suggest that new compounds affecting the NPR-1 pathway are still worth investigating.

STUDY LIMITATIONS. First, only 30% of the study population was female, so the impact of NPR-1 agonism on sex cannot be fully characterized. Second, we did not explore a dose >240 mg, and it is possible that a higher dose of the NPR-1 agonist might have had greater antihypertensive efficacy. However, the prior phase 1 study[6] showed that doses <240 mg lowered BP significantly, and we did not want to risk excessive hypotension in this more complicated patient population. Adherence to background antihypertensive therapy was only 55% in our patient population, but we have no evidence that this led to the neutral effect on ambulatory BP of XXB750 nor the substantial reduction in ambulatory BP with placebo. The administration of XXB750 was parenteral, there was a significant increase in drug concentration with dose, and the impact on cGMP by dose was robust. Finally, there was a nominal increase in AEs in patients randomized to receive XXB750 compared with placebo, including 3 cases of pneumonia. Although there is no known mechanism for increases in respiratory infections with NPR-1 agonism, the sample size in this phase 2 study was too small to determine incidence rates of AEs with precision.

CONCLUSIONS

In summary, the current study noted dose-related changes in cGMP supporting target engagement of XXB750 at the NPR-1 receptor. However, there were no dose-response patterns observed in change from baseline in 24-hour SBP, with none of the doses of XXB750 having a greater BP-lowering effect than placebo. XXB750 was well tolerated; most AEs were resolved spontaneously, and no dose-related trends in the incidence rates were noted.

XXB750 did not lead to a significant reduction in BP in patients with resistant hypertension at doses up to 240 mg once monthly despite a clear engagement of the secondary messenger cGMP. Overall, the neutral results of this first-in-class trial of an NPR-1 agonist in resistant hypertension highlight the importance of continued research and development of new antihypertensive treatments to address the unmet needs of patients with resistant hypertension. Importantly, there is substantial value in a well-conducted trial even when the results are neutral. We believe that more research is necessary to better understand the clinical pharmacology of XXB750 and its potential effects on other biological systems within hypertensive disease models. Alternative strategies or modalities that ensure more consistent activation of the NPR-1 receptors (eg, small molecules with improved renal tubule accessibility) should also be explored to enhance the effectiveness of natriuretic peptide system modulation.

DATA AVAILABILITY The methods and data sets generated and analyzed during the current study are not publicly available. Novartis is committed to sharing access to patient-level data and supporting clinical documents from eligible studies with qualified external researchers. These requests are reviewed and approved based on scientific merit. All data provided are anonymized to respect the privacy of patients who have participated in the trial, in line with applicable laws and regulations. The data can be requested from the corresponding author of the article. The protocol can be made available on request by contacting the corresponding author.

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KEY WORDS blood pressure, cyclic GMP, NPR-1 agonist, resistant hypertension, XXB750

APPENDIX For supplemental methods, figures, and tables, please see the online version of this paper.